Three deep-learning priors for image dehazing

ROAS 5800 Assignment 2 · Yi Zhang · Student ID 50063117 · 7 November 2026

1. Problem and survey scope

Single-image dehazing estimates a clear RGB scene from one hazy observation. The atmospheric model I(x) = J(x)t(x) + A[1 − t(x)] illustrates the ambiguity: scene radiance J, transmission t and airlight A are unknown. Prior-based methods constrain these quantities; learned restoration uses CNN/attention baselines such as FFA-Net and DehazeFormer, frequency information or generative priors [1–3]. This survey examines SFMN, SFHformer and IPC-Dehaze, followed by a controlled laptop deployment test. Their publication-time performance claims concern different benchmarks and objectives, rather than a universal ranking. An accompanying UI supports new-image inference and saved comparisons, with separate Assignment 1/2 website routes.

2. Motivations, mechanisms and drawbacks

SFMN [1] addresses incomplete context aggregation with cross-scale and cross-level feature modulation. A two-stage U-Net first restores Fourier amplitude information, then refines phase-related structure. A Cross-Scale Modulator combines resolutions, a Frequency Modulator processes spectral features, and a Cross-Level Modulator exchanges hierarchical information. This avoids a large attention matrix, but two restoration stages and feature exchange add work. Transfer from synthetic training to real haze remains an empirical question.

SFHformer [2] targets the complementary roles of local texture and global degradation. Its Local-Global Perception Mixer combines a spatial branch with an FFT branch using frequency-conditioned position encoding and dynamic convolution. Multi-kernel ConvFFN adds local receptive fields. FFT and dynamic operations still incur runtime cost, and the learned priors depend on training data. The strictly loaded official dehazing release contains 4.75M parameters, whereas the paper table lists 3.87M; the local test evaluates the release configuration and does not reproduce that table.

IPC-Dehaze [3] addresses errors in one-shot clean-code prediction with a pretrained VQGAN codebook. Trained using synthetic pairs, it encodes input features; Code-Predictor proposes discrete codes, and Code-Critic evaluates which codes to resample. Eight iterative rounds precede decoding, while spatial feature transforms preserve scene information. This supplies a high-quality generative prior but increases latency and can change image detail. Its perceptual evaluation objective differs from strict pixel fidelity.

3. Results reported by the papers

Method	Original evaluation	Paper-reported score
SFMN [1]	SOTS-Outdoor, PSNR / SSIM	37.72 dB / 0.991
SFHformer [2]	SOTS-Outdoor, PSNR / SSIM	38.83 dB / 0.9951
IPC-Dehaze [3]	RTTS, MUSIQ / TOPIQ	59.60 / 0.500

Sources: [1] Table I, [2] Table 1, [3] Table 1. Higher is better for all listed metrics. RTTS no-reference perceptual scores and SOTS reference scores measure different properties and cannot be combined into one leaderboard. Original figures and additional background appear in the slides and companion guide.

4. Complete evaluation on the RTX 4060 laptop

All 352 original RW2AH test pairs [4] were evaluated at 640×512 without resizing. Source hashes and pairing were verified. Hardware was an RTX 4060 Laptop GPU (8GB), Ryzen 9 7845HX and 15.2GiB usable RAM, running PyTorch 2.8.0+cu128. Official OTS weights were used for SFMN/SFHformer and the official real-world release for IPC-Dehaze. All models ran sequentially at batch 1 in FP32 with TF32 disabled. Common mirror padding was removed before PNG export. Both SFMN stages and all eight IPC rounds were retained.

Method	PSNR (dB)	SSIM	GPU time (ms)	Peak (MiB)
Hazy input	18.62	0.5564	—	—
SFMN	18.32	0.5561	241.3	1553
SFHformer	18.37	0.5553	505.2	586
IPC-Dehaze	13.21	0.4918	5379.7	849

Quality: equal-weight mean of per-image RGB PSNR and Gaussian-window SSIM on saved uint8 PNGs. Time: dataset median of per-image medians from three complete GPU forwards after warmup; model loading, transfers and image I/O are excluded. Memory: maximum PyTorch allocated-tensor peak, including model/input tensors. All 1,056 outputs and their quality metrics were independently verified.

All three releases reduced mean PSNR relative to the hazy-input baseline. SFHformer obtained the highest restored-image mean PSNR (18.37 dB), 0.25 dB below the hazy-input baseline (18.62 dB). SFMN had the lowest median GPU forward time (241.3 ms). All three completed the full test within 8GB device capacity.



Figure 1. Native-resolution outputs for RW2AH pair 8-0121-02. The case was selected at the 75th-percentile rank of input PSNR, independently of model rankings. The images illustrate scene-dependent changes; the table uses the full test set.

5. Interpretation and future work

The measured scores test real-haze transfer and deployment, not equal-budget training or the papers' original benchmark protocols. Training distributions differ, and reference metrics can penalize color/detail changes that improve perceived clarity. A single dataset cannot establish universal superiority or explain the cause of each failure. Further work should use several real-haze datasets, inspect video consistency and test downstream detection. No detector gain or training-from-scratch result is claimed.

References

[1] H. Shen, H. Ding, Y. Zhang, Z.-Q. Zhao and X. Jiang. Spatial Frequency Modulation Network for Efficient Image Dehazing. IEEE Transactions on Image Processing 34, 2025.

[2] X. Jiang, X. Zhang, N. Gao and Y. Deng. When Fast Fourier Transform Meets Transformer for Image Restoration. ECCV 2024, 381–402.

[3] J. Fu et al. Iterative Predictor-Critic Code Decoding for Real-World Image Dehazing. CVPR 2025, 12700–12709.

[4] W. Fang et al. Guided Real Image Dehazing Using YCbCr Color Space. AAAI 39(3), 2906–2914, 2025. doi:10.1609/aaai.v39i3.32297.